

Quantum-Inspired AI: Multi-stage Reasoning Research Analysis & Improvement Roadmap

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Executive Summary

This report provides a comprehensive literature-driven analysis of the Samsung PRISM project "**Quantum-inspired Annealing for Multi-stage Reasoning using Small Language Models (SLMs)**". The project aims to design a pipeline that couples quantum-inspired QUBO optimization with SLMs (Llama-3.2-3B+) to improve multi-step reasoning accuracy while targeting a **>2× performance gain** over the baseline Llama-3.2-3B model. The following sections survey ten key published works, compare methodologies and benchmark results, identify gaps and improvement opportunities, and provide concrete suggestions to achieve the 2× target.

1. Project Pipeline Overview

The pipeline consists of three stages: (i) **Sampling** — the SLM generates N answers $\times$ M reasons each from an initial prompt; (ii) **QUBO Optimization** — all reasons are mapped to a QUBO matrix and solved by a quantum / quantum-inspired annealer to select the best subset; (iii) **Final Inference** — the selected reasons plus the query form a final prompt fed to the SLLM for the final answer. The approach combines Chain-of-Thought (CoT) prompting philosophy with combinatorial optimization, inspired primarily by Esencan et al. (2024) and extended towards quantum hardware by Flores-Garrigos et al. (2025).

2. Comparative Analysis of Key Related Publications

The table below compares 10 core papers spanning reasoning methods, quantum-classical optimization, and SLM benchmarking. Column explanations: **Model/Scale** = primary model used; **Reasoning Method** = how reasoning is structured; **Optimizer** = how answers/paths are selected; **Benchmark** = evaluation dataset; **Best Accuracy / Score** = reported result; **Gap / Limitation** = identified weakness; **Relevance to PRISM** = how it informs this project.

Paper / Year	Model / Scale	Reasoning Method	Optimizer	Benchmark	Best Score	Gap / Limitation
Wei et al. (2022) Chain-of-Thought Prompting [NeurIPS 2022]	PaLM 540B, GPT-3 175B	Few-shot CoT: step-by-step reasoning chain appended to prompt	Greedy Decoding	GSM8K, SVAMP, AQuA, CommonsenseQA	GSM8K: 56.5% (PaLM 540B)	Requires large model ($\geq 100B$); limited diversity; single reasoning path
Wang et al. (2023) Self-Consistency [ICLR 2023]	PaLM 540B, GPT-3 175B	Multiple diverse CoT paths; majority voting	Majority Vote	GSM8K, SVAMP, AQuA, ARC	GSM8K: +17.9% over baseline CoT	Majority vote suboptimal for non-symmetric answers; high sampling cost ($\times 40$ calls)
Yao et al. (2023) Tree of Thoughts [NeurIPS 2023]	GPT-4 / GPT-3.5	Tree search over partial thoughts; BFS/DFS with LLM evaluator	BFS / DFS + LLM Heuristic	Game of 24, Creative Writing, Mini Crosswords	Game-of-24: 74% vs 4% (IO)	Exponential search cost; not evaluated on SLMs; no external optimizer
Yao et al. (2022) ReAct [ICLR 2023]	PaLM 540B	Synergizes reasoning + acting (tool calls) interleaved	Greedy + Tool Feedback	HotpotQA, FEVER, AlfWorld, WebShop	HotpotQA: 35.1% EM (QA tasks)	Relies on external API/tool; latency high; no combinatorial selection

Paper / Year	Model / Scale	Reasoning Method	Optimizer	Benchmark	Best Score	Gap / Limitation
Esencan et al. (2024) Combinatorial Reasoning (CR) [arXiv 2024]	GPT-3.5 / GPT-4 (Large LLMs)	Sample N reasons; map to QUBO; select best subset	QUBO + Ising Machine (Fujitsu DA)	BigBench-Hard (BBH)	>12% over zero-shot; near human-level on BBH tasks	Only large LLMs tested; QUBO with avg 900 vars; no SLM adaptation; no 2× target
Flores-Garrigos et al. (2025) QCR-LLM [arXiv Oct 2025]	Llama-3 8B, GPT-4o	Higher-order HUBO/QUBO over reasoning fragments with quantum solver	BF-DCQO Quantum Annealer	BIG-Bench Extra Hard (BBEH)	Surpasses reasoning-native systems on BBEH	Hardware-dependent (quantum device needed); higher-order terms hard to scale on NISQ devices
Margapuri et al. (2025) PEPS + PPO [arXiv Sep 2025]	TinyLLaMA-1.1B (SLM)	PEPS fidelity reward in PPO for coherent reasoning traces	Proximal Policy Optimization	GSM8K, StrategyQA, EntailmentBank	GSM8K MEC: +10%, EntailmentBank MEC: +13.6%	Requires RL fine-tuning; coherence metric not accurate; no QUBO integration
Wang et al. (ACL 2025) Ranked Voting Self-Consistency	Various LLMs	Ranked voting over sampled reasoning paths instead of simple majority	Ranked Voting (Borda count / IRV)	Multiple reasoning benchmarks	Outperforms majority voting, especially on complex tasks	No combinatorial optimizer; voting still heuristic; no SLM-specific tuning
Kojima et al. (2022) Zero-Shot CoT [NeurIPS 2022]	GPT-3 175B, PaLM 540B	'Let's think step by step' zero-shot prompt trigger	Greedy Decoding	MultiArith, GSM8K, SVAMP, AQuA, SingleEQ	MultiArith: 78.7% (GPT-3)	Needs large model; no diverse sampling; single path; no optimization
Chandarana et al. (2025) Runtime Quantum Advantage [arXiv 2025]	Digital Quantum Processor	Counterdiabatic quantum optimization for QUBO / Ising models	BF-DCQO (Digital Quantum)	Benchmark QUBO problems	Runtime advantage over classical SA on structured problems	Requires quantum hardware; not yet integrated with LLM pipeline; NISQ limitations

Table 1: Comparative analysis of key related publications. DA = Digital Annealer, QUBO = Quadratic Unconstrained Binary Optimization, BBH = BigBench-Hard, BBEH = BIG-Bench Extra Hard, MEC = Multi-step Entailment Coherence.

3. Benchmark Performance Reference Table

To understand the 2× performance target, we first establish baseline and SOTA scores on the most relevant benchmarks used in multi-step reasoning research. The table below covers GSM8K (grade-school math), BBH / BBEH (complex reasoning), StrategyQA (commonsense), and MMLU (general knowledge). The **Llama-3.2-3B baseline** is the project's starting point; the **2× target** refers to doubling the effective reasoning accuracy gain over the baseline.

Model / Method	GSM8K Accuracy	BBH / BBEH Score	StrategyQA Acc.	MMLU Score	Notes
Llama-3.2-3B (Baseline, maj@1)	~58–62%	~42–45%	~62%	~63%	Baseline SLM used in this project (3B params, pruned/distilled from Llama-3 8B)
Llama-3.2-3B + CoT (few-shot)	~65–68%	~48%	~65%	~64%	CoT prompting adds ~5–8% on math; standard baseline for prompting approaches
Llama-3.2-3B + Self-Consistency (SC, k=40)	~72–75%	~54%	~68%	~65%	Wang et al. 2023; majority vote over 40 paths; high compute overhead
Llama-3.1-8B + CoT (maj@1)	~80.1%	~57%	~72%	~84.6%	Upper-bound reference for 8B scale; PRISM targets matching this with 3B + QUBO
Phi-3.5-mini (3.8B) + CoT	~86.2%	~60%	~74%	~69%	Strongest competing SLM on GSM8K; highly optimized for math reasoning
Esencan CR (GPT-3.5) + QUBO (Fujitsu DA)	N/A (BBH focus)	>12% gain over zero-shot	N/A	N/A	Proof-of-concept on large LLMs; QUBO with ~900 vars; not adapted for SLMs
QCR-LLM (Flores-Garrigos 2025) + Quantum Annealer	N/A	Surpasses reasoning-native models on BBEH	N/A	N/A	Higher-order QUBO (HUBO); requires quantum hardware; state-of-art on BBEH
PEPS+PPO (TinyLLaMA-1.1B) Quantum-inspired RL	MEC +10% over baseline	N/A	Coherence metric +7%	N/A	Coherence-focused; not accuracy; integrating QUBO selection would further improve
2× TARGET (PRISM Goal) Llama-3.2-3B + QA-Pipeline	>90% (target)	>80% (target)	>80% (target)	>70% (target)	2× means doubling the accuracy GAIN over baseline greedy: e.g., +20% gap → +40% gap

Table 2: Benchmark reference scores across methods and models. N/A = not reported. Green row = project 2× target. GSM8K scores are accuracy %; BBH/BBEH are task-specific scores; MEC = Multi-step Entailment Coherence (0–1). Llama-3.2-3B baseline scores estimated from Meta model card and comparative literature.

4. Research Gap Analysis: Where Can We Improve?

After examining all publications, the following improvement areas are identified. Each area includes the current state in related work, the gap relative to PRISM objectives, the potential gain, and a concrete improvement strategy.

Improvement Area	Current State (Literature)	Gap vs PRISM Target	Est. Gain	Proposed Improvement Strategy
1. QUBO Formulation for SLMs	Esencan 2024 uses ~900 QUBO variables on LLMs. QCR-LLM extends to HUBO but needs quantum hardware.	No QUBO formulation tested on 3B SLMs. Sparse reasons from small models may degrade QUBO quality.	+15–25% acc. gain	Design a lightweight QUBO with <200 binary variables by clustering semantically similar reasons before mapping. Use TF-IDF or sentence embeddings to reduce variable count while preserving diversity. Test SA (simulated annealing) as the solver on CPU first.
2. Diversity of Sampled Reasoning Paths	Self-consistency (Wang 2023) uses temperature sampling. ToT uses tree search. Both designed for large models.	SLMs at 3B generate lower-diversity outputs at high temperature. Majority vote unreliable for SLMs (MV errors are correlated).	+10–18% acc. gain	Apply contrastive decoding or nucleus sampling with adaptive temperature per token. Combine with prompt-level perturbation (rephrased queries) to generate structurally diverse reasoning paths before QUBO selection.
3. Reason Quality Scoring in QUBO	Esencan 2024 uses pairwise similarity as QUBO weights. QCR-LLM uses fragment stability scores.	Pairwise similarity ignores factual correctness. SLM-generated reasons often contain hallucinations that poison the QUBO objective.	+12–20% acc. gain	Add a lightweight verifier (rule-based for math; small NLI model for commonsense) to score reason plausibility before building the QUBO matrix. Include correctness score as a diagonal term in the QUBO Q-matrix.
4. Hyperparameter Optimization via Quantum-inspired Annealing	Llama-3.2-3B uses fixed hyperparameters. Standard grid search is computationally expensive.	No published work combines QUBO-based hyperparameter search with multi-stage reasoning SLM pipelines.	+8–15% latency / acc.	Encode key inference hyperparameters (temperature, top-p, num_beams, num_samples N, reason subset size K) as binary variables in a QUBO. Use simulated annealing on the Nvidia A100 to find the optimal configuration jointly, rather than sequential grid search.
5. Higher-Order QUBO (HUBO) Integration	QCR-LLM (Flores-Garrigos 2025) uses HUBO for triple-wise reason interactions. Standard QUBO is quadratic.	Pairwise QUBO misses three-way or higher reasoning dependencies. HUBO not yet tested on SLMs.	+5–10% acc. gain	Extend the QUBO matrix to include triple-wise reason interaction terms (cubic terms). Use polynomial unconstrained binary optimization (PUBO) reducible to QUBO via auxiliary variables. Limit to triplets of highest-confidence reasons to keep problem size tractable on CPU/A100.
6. Context Window Utilization in Final Prompt	Pipeline feeds selected reasons + query as final prompt. Llama-3.2-3B has 128K context but typical prompts use <2K tokens.	Suboptimal use of context. Selected reasons may be redundant or improperly ordered. No ranking of reasons in final prompt.	+5–8% acc. gain	After QUBO selects the K best reasons, re-rank them by relevance to the query using a cosine-similarity ordering. Prepend the most relevant reasons closest to the final question. Experiment with 3–8 reasons to find the sweet spot without context overflow.

Improvement Area	Current State (Literature)	Gap vs PRISM Target	Est. Gain	Proposed Improvement Strategy
7. Model Fine-Tuning with QUBO-Selected Traces	PEPS+PPO (Margapuri 2025) fine-tunes TinyLLaMA with RL rewards. Current PRISM pipeline is inference-only.	No RL or SFT fine-tuning step in current PRISM design. Fine-tuning could solidify improvements from QUBO selection.	+10–20% acc. gain	Use QUBO-selected high-quality reasoning traces as supervised fine-tuning (SFT) data. Run 2–3 epochs of SFT on the Llama-3.2-3B model using A100 GPUs. This creates a reinforcement loop where the model learns to self-generate QUBO-friendly diverse reasoning paths.
8. Simulated Annealing Schedule Optimization	D-Wave and digital annealers use fixed schedules. Chandarana 2025 shows counterdiabatic schedules improve convergence.	Vanilla simulated annealing on CPU may converge to local optima for large QUBO problems (>500 vars). Suboptimal annealing = wrong reason selection.	+5–12% QUBO quality	Implement counterdiabatic-inspired annealing schedules (BF-DCQO analog) in simulation on GPU. Add momentum terms to the SA update rule to escape local minima. Compare against vanilla SA and Tabu search to identify best CPU/GPU solver for the QUBO size generated by 3B SLMs.
9. Multi-Benchmark Evaluation	Esencan 2024 uses only BBH. QCR-LLM uses BBEH. PEPS uses GSM8K, StrategyQA, EntailmentBank.	PRISM lacks a cross-benchmark evaluation plan. The 2× claim needs to be validated across diverse tasks.	Validation completeness	Evaluate on at minimum: GSM8K (math), BBH (complex reasoning), StrategyQA (commonsense), ARC-Challenge (science). Report accuracy, reasoning trace coherence (MEC, WES), and throughput (tokens/sec on A100). This enables apples-to-apples comparison with all baseline papers.
10. Compute Efficiency / Latency Reduction	Self-consistency requires 40 LLM calls. ToT requires tree-depth × branching factor calls. Both impractical for SLMs on edge devices.	QUBO-based selection also requires N×M samples. On 3B SLMs with A100, total FLOPS budget must be managed to stay within 2× latency.	Latency 2× target	Use early-stopping in the sampling stage: stop sampling once QUBO objective converges (measured by successive low-change iterations). Target N=5 answers × M=5 reasons = 25 paths, down from 40 in self-consistency, while recovering quality through QUBO selection quality.

Table 3: Research gap analysis and improvement strategies. 'Est. Gain' refers to expected improvement over the Llama-3.2-3B CoT baseline in accuracy or efficiency. Strategies are ordered by estimated impact on achieving the 2× performance benchmark.

5. Path to 2× Performance: Compounded Improvement Strategy

The 2× performance gain target means that the accuracy improvement from the pipeline must be at least double that achievable by the SLM alone. For example, if Llama-3.2-3B achieves 62% on GSM8K (greedy) and 68% with basic CoT (a +6% gain), the pipeline must achieve at least $62\% + 2 \times 6\% = 74\%$ or above. More ambitiously, doubling the absolute accuracy from ~62% to $\geq 90\%$ on GSM8K requires a compounded multi-faceted approach. The strategies below are stacked in order of implementation priority.

Priority	Strategy	Description	Expected Impact	Phase (Roadmap)	Resources Needed
P1	Improved QUBO Mapping with Reason Verifier	Build lightweight verifier (NLI or rule-based) to score SLM reasons before QUBO mapping. Include correctness scores as diagonal QUBO terms. Reduces noise from hallucinated reasons.	GSM8K: +15–20%; BBH: +12–15%	Jun–Jul 2026	A100 GPU; small NLI model (RoBERTa)
P2	Diverse Sampling with Contrastive Decoding	Apply prompt perturbation + adaptive temperature to maximize reason diversity. Fewer but higher-quality diverse paths improves QUBO selection.	GSM8K: +8–12%; StrategyQA: +10%	Jun–Jul 2026	A100 GPU; HuggingFace transformers
P3	SFT on QUBO-Selected Reasoning Traces	Collect QUBO-selected high-quality traces from sampling; fine-tune Llama-3.2-3B for 2–3 epochs. Model learns to self-generate better reasoning aligned with QUBO selection.	GSM8K: +10–15%; MMLU: +5–8%	Jul–Aug 2026	A100 GPU ×4; 10K labeled examples
P4	Hyperparameter QUBO Search	Encode inference hyperparameters as binary QUBO variables. Joint optimization replaces sequential grid search, finding globally optimal sampling configuration.	Latency: –30%; Accuracy: +5–8%	Jun–Jul 2026	A100 GPU; D-Wave Ocean SDK or PyQUBO
P5	HUBO Extension for 3-way Reason Dependencies	Extend QUBO to cubic terms for triplet reason interactions. Reduces to QUBO via auxiliary variables. Captures complex logical chains missing in pairwise QUBO.	BBH/BBEH: +5–10%	Aug–Sep 2026	PyQUBO / dimod; A100 GPU
P6	Optimized Annealing Schedule (BF-DCQO inspired)	Implement counterdiabatic-inspired momentum SA on GPU. Escape local minima better than vanilla SA. Benchmarked against Tabu search and D-Wave simulator.	QUBO solution quality: +8–12%	Aug–Sep 2026	CUDA-compatible SA; A100 GPU
P7	Multi-Benchmark Evaluation Suite	Systematic evaluation on GSM8K, BBH, StrategyQA, ARC-Challenge, MMLU. Track accuracy, coherence (MEC/WES), throughput, and QUBO convergence time.	Validates 2× claim; publishable results	Aug–Sep 2026	A100 GPU; evaluation scripts

Table 4: Prioritized implementation roadmap to achieve 2× performance. P1–P7 ordered by impact and feasibility within the Jun–Sep 2026 timeline.

6. Estimated Cumulative Performance Gain

The following table projects the expected cumulative accuracy on GSM8K as each strategy is applied incrementally on Llama-3.2-3B. Values are conservative estimates based on published results from the comparable papers and scaled for the 3B parameter regime.

#	Stage Added	Method / Reference	GSM8K Acc. (Estimated)	Gain over Previous Stage	2× Target Progress
0	Llama-3.2-3B Greedy Baseline	Meta model card	~60%	—	0%
1	+ CoT Few-shot Prompting	Wei et al. 2022	~66%	+6%	~15%
2	+ Diverse Sampling (N=5, M=5)	Wang et al. 2023 (adapted)	~72%	+6%	~40%
3	+ QUBO Reason Selection (baseline CR)	Esencan et al. 2024	~78%	+6%	~60%
4	+ Reason Verifier in QUBO (P1)	Novel PRISM contribution	~84%	+6%	~80%
5	+ SFT on QUBO Traces (P3)	PEPS-inspired, PRISM	~88%	+4%	~93%
6	+ Optimized Annealing + HUBO (P5, P6)	QCR-LLM + Chandarana 2025 inspired	~91%	+3%	~103% (2× ACHIEVED)

Table 5: Projected cumulative GSM8K accuracy as pipeline components are added to Llama-3.2-3B. 2× target progress = (current gain) / (2 × baseline CoT gain of ~6%). Final stage exceeds 2× target. Values are estimates; actual results depend on implementation quality.

7. Novel Research Contributions Proposed for PRISM

7.1 SLM-Adapted Lightweight QUBO Framework

Existing CR (Esencan 2024) and QCR-LLM (Flores-Garrigos 2025) were designed for large LLMs (GPT-3.5+). A key novel contribution of PRISM is adapting the QUBO formulation for the lower-quality, lower-diversity outputs of 3B SLMs. This involves semantic clustering before QUBO mapping to reduce the variable space from ~900 variables (Esencan) to ~100–200 variables, making it tractable on CPU/A100 without quantum hardware.

7.2 Correctness-Aware QUBO Objective

All prior works use similarity-based QUBO weights. PRISM can pioneer the inclusion of a correctness/plausibility score (from a lightweight verifier) as a bonus term in the QUBO diagonal. This directly biases the optimizer towards factually sound reason subsets — a capability not present in any of the cited papers — and is likely the single highest-impact improvement for math reasoning.

7.3 Joint Hyperparameter and Reason Selection QUBO

No existing paper jointly optimizes SLM inference hyperparameters and reason selection in a single QUBO. By encoding both as binary variables, PRISM can find the globally optimal configuration in one annealing run, which is a methodologically novel and potentially publishable contribution.

7.4 QUBO-Guided SFT Feedback Loop

Inspired by PEPS+PPO (Margapuri 2025), but replacing RL with simpler SFT, PRISM can create a feedback loop: QUBO selects the best reasoning traces → these become SFT training data → the fine-tuned SLM generates better initial reasons → QUBO quality improves further. This iterative self-improvement loop is not explored in any of the surveyed papers.

8. Conclusion

The Samsung PRISM project on Quantum-inspired Annealing for Multi-stage Reasoning is well-positioned at the intersection of two active research fronts: advanced LLM reasoning and combinatorial optimization. The literature analysis reveals that while QUBO-based reason selection (Esencan 2024, QCR-LLM 2025) has shown promise on large LLMs, no work has systematically applied and optimized this paradigm for small language models under resource constraints.

The most impactful improvements identified are: (1) adding a correctness-aware verifier to the QUBO objective function, (2) diversifying SLM sampling before QUBO mapping, (3) iterative SFT on QUBO-selected traces, and (4) a joint hyperparameter+reason-selection QUBO. Applied together with the roadmap in Table 4, these strategies are projected to achieve >90% on GSM8K — exceeding the 2× performance gain target — while maintaining feasibility on the Nvidia A100 infrastructure planned for the Jun–Sep 2026 phase.

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